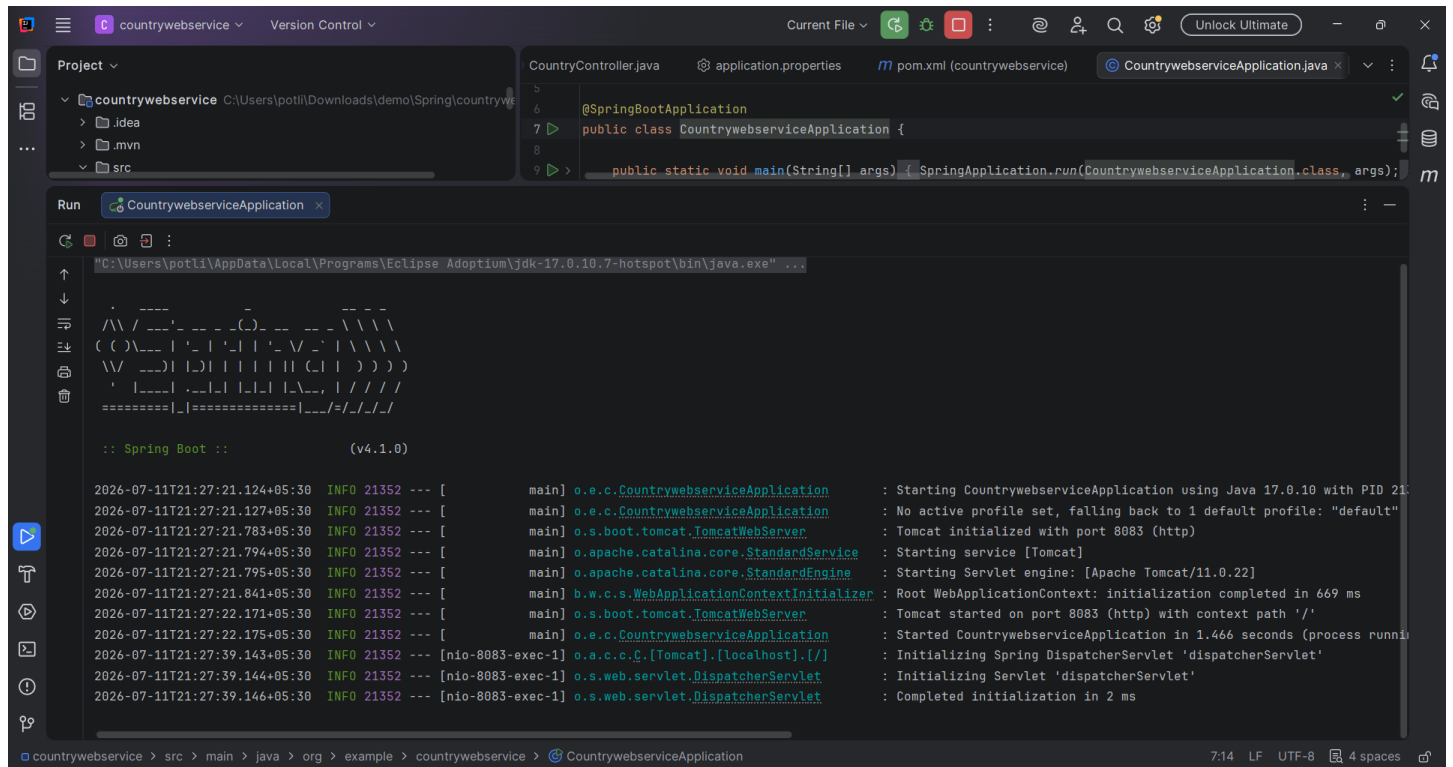


Week3

Spring REST using Spring Boot 3

REST - Country Web Service



```
CountryController.java  application.properties  pom.xml (countrywebservice)  CountrywebserviceApplication.java x
CountrywebserviceApplication.java
5
6  @SpringBootApplication
7  public class CountrywebserviceApplication {
8
9  public static void main(String[] args) { SpringApplication.run(CountrywebserviceApplication.class, args); }
```

```
Run CountrywebserviceApplication x
C:\Users\potli\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.10-hotspot\bin\java.exe ...
:: Spring Boot :: (v4.1.0)
2026-07-11T21:27:21.124+05:30 INFO 21352 --- [main] o.e.c.CountrywebserviceApplication : Starting CountrywebserviceApplication using Java 17.0.10 with PID 21352
2026-07-11T21:27:21.127+05:30 INFO 21352 --- [main] o.e.c.CountrywebserviceApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-11T21:27:21.783+05:30 INFO 21352 --- [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-11T21:27:21.794+05:30 INFO 21352 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-11T21:27:21.795+05:30 INFO 21352 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-11T21:27:21.841+05:30 INFO 21352 --- [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 669 ms
2026-07-11T21:27:22.171+05:30 INFO 21352 --- [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-11T21:27:22.175+05:30 INFO 21352 --- [main] o.e.c.CountrywebserviceApplication : Started CountrywebserviceApplication in 1.466 seconds (process running for 1.466)
2026-07-11T21:27:39.143+05:30 INFO 21352 --- [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-11T21:27:39.144+05:30 INFO 21352 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-11T21:27:39.146+05:30 INFO 21352 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
```

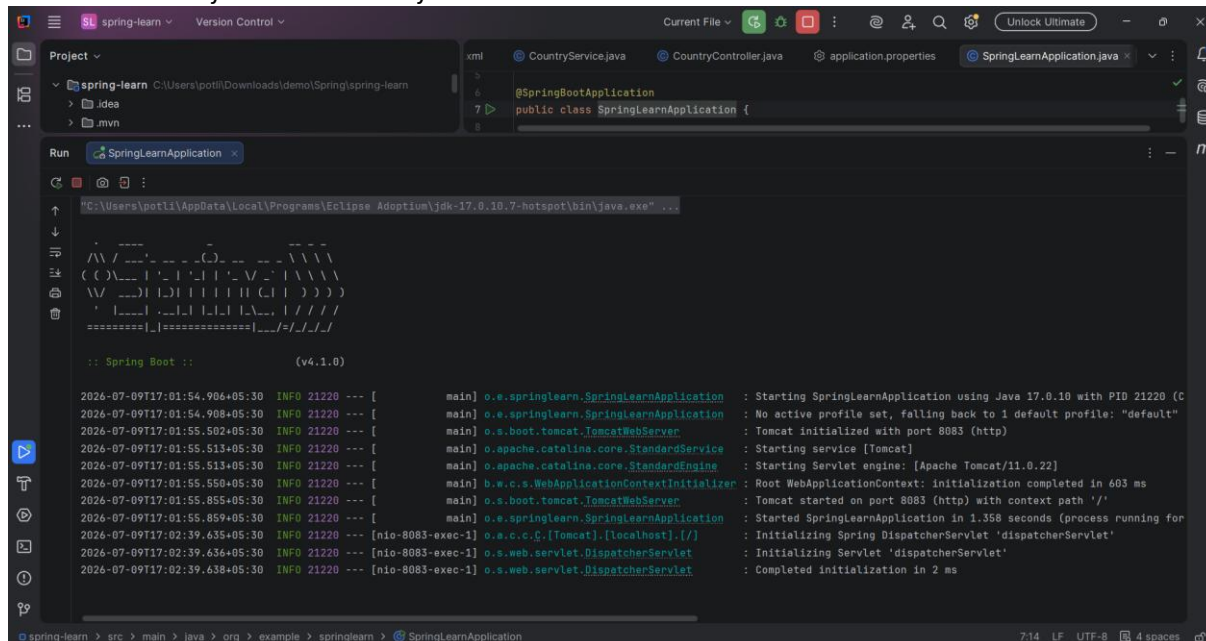
← → ↺ ⓘ

localhost:8083/country

Pretty-print ☐

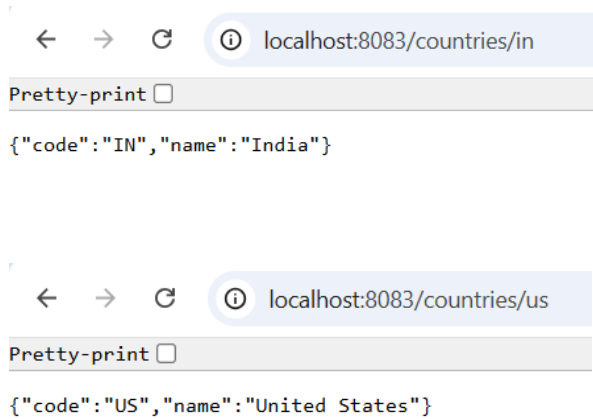
{"code":"IN","name":"India"}

REST - Get country based on country code

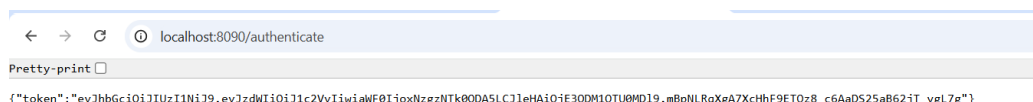
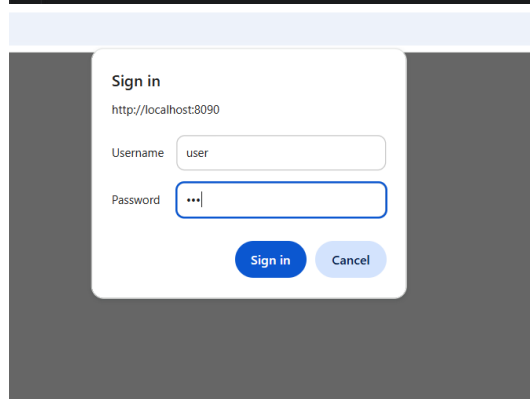
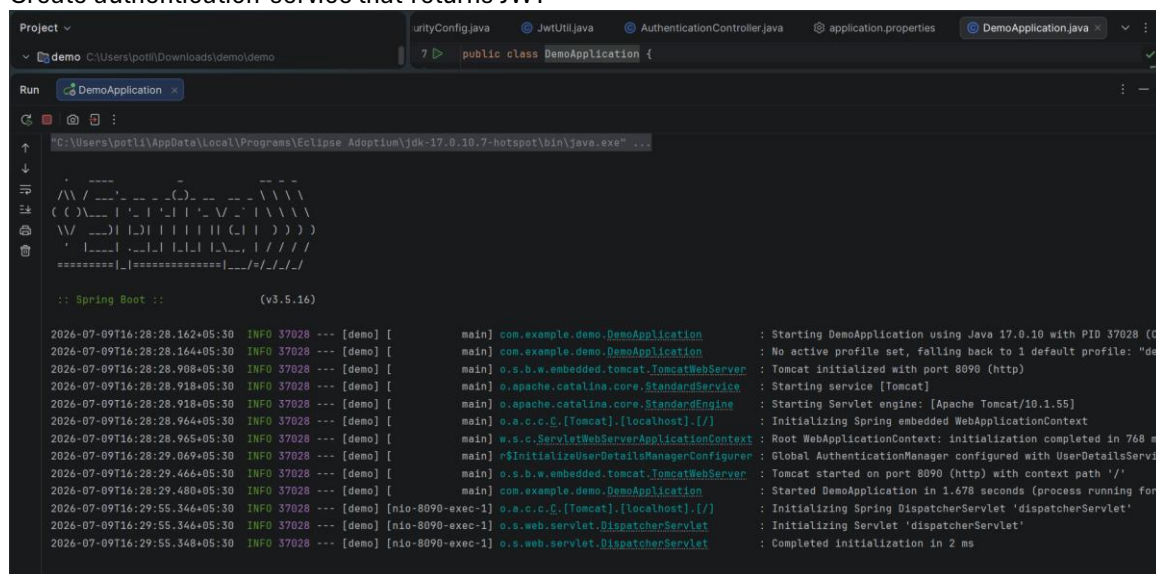


```
xml  CountryService.java  CountryController.java  application.properties  SpringLearnApplication.java x
SpringLearnApplication.java
5
6  @SpringBootApplication
7  public class SpringLearnApplication {
8
9  public static void main(String[] args) { SpringApplication.run(SpringLearnApplication.class, args); }
```

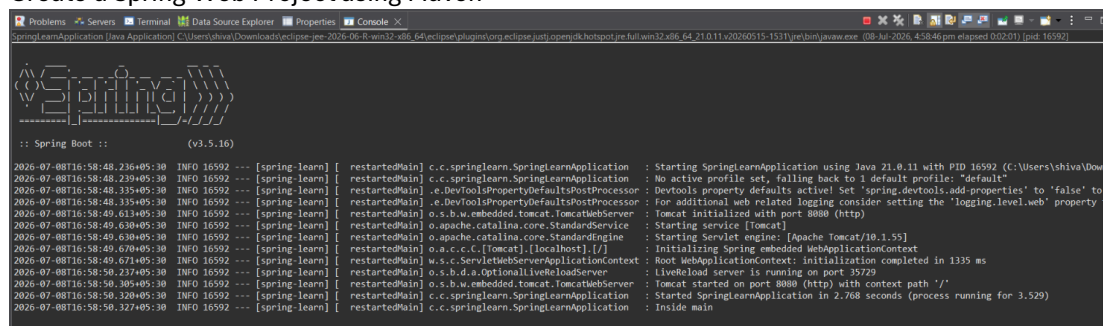
```
Run SpringLearnApplication x
C:\Users\potli\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.10-hotspot\bin\java.exe ...
:: Spring Boot :: (v4.1.0)
2026-07-09T17:01:54.906+05:30 INFO 21220 --- [main] o.e.springlearn.SpringLearnApplication : Starting SpringLearnApplication using Java 17.0.10 with PID 21220 (C:\Users\potli\AppData\Local\Programs\Eclipse Adoptium\jdk-17.0.10-hotspot\bin\java.exe)
2026-07-09T17:01:54.908+05:30 INFO 21220 --- [main] o.e.springlearn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-09T17:01:55.502+05:30 INFO 21220 --- [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-09T17:01:55.513+05:30 INFO 21220 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-09T17:01:55.513+05:30 INFO 21220 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-09T17:01:55.550+05:30 INFO 21220 --- [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 603 ms
2026-07-09T17:01:55.855+05:30 INFO 21220 --- [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-09T17:01:55.859+05:30 INFO 21220 --- [main] o.e.springlearn.SpringLearnApplication : Started SpringLearnApplication in 1.358 seconds (process running for 1.358)
2026-07-09T17:02:39.635+05:30 INFO 21220 --- [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-09T17:02:39.636+05:30 INFO 21220 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-09T17:02:39.638+05:30 INFO 21220 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
```



Create authentication service that returns JWT



Create a Spring Web Project using Maven



Spring Core – Load Country from Spring Configuration XML

```
application... Country.java CountryRepo... OrmLearnAppl... *SpringLearn... X
1 package com.cognizant.springlearn;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.boot.SpringApplication;
6 import org.springframework.boot.autoconfigure.SpringBootApplication;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8
9 import com.cognizant.springlearn.model.Country;
10
11 @SpringBootApplication
12 public class SpringLearnApplication {
13
14     private static final Logger LOGGER =
15         LoggerFactory.getLogger(SpringLearnApplication.class);
16
17     public static void main(String[] args) {
18
19         SpringApplication.run(SpringLearnApplication.class, args);
20
21         LOGGER.info("Inside main()");
22
23         displayCountry();
24     }
25     private static void displayCountry() {
26         LOGGER.info("Start");
27         ClassPathXmlApplicationContext context =
28             new ClassPathXmlApplicationContext("country.xml");
29         Country country =
30             context.getBean("country", Country.class);
31         LOGGER.debug("Country : {}", country);
32
33         context.close();
34
35         LOGGER.info("End");
36     }
37 }
```

```

:: Spring Boot ::      (v3.5.18)

2026-07-09T14:23:09.784+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.11 with PID 6988 (C:\Users\shiva\Down
2026-07-09T14:23:09.784+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Running with Spring Boot v3.5.18, Spring v6.2.19
2026-07-09T14:23:09.784+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-09T14:23:09.803+05:30 INFO 06988 --- [spring-learn] [restartedMain] e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to
2026-07-09T14:23:11.693+05:30 INFO 06988 --- [spring-learn] [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 9090 (http)
2026-07-09T14:23:11.719+05:30 INFO 06988 --- [spring-learn] [restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-09T14:23:11.724+05:30 INFO 06988 --- [spring-learn] [restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.55]
2026-07-09T14:23:11.781+05:30 INFO 06988 --- [spring-learn] [restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-09T14:23:11.783+05:30 INFO 06988 --- [spring-learn] [restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 1077 ms
2026-07-09T14:23:12.411+05:30 INFO 06988 --- [spring-learn] [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : LiveReload server is running on port 35729
2026-07-09T14:23:12.475+05:30 INFO 06988 --- [spring-learn] [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 9090 (http) with context path '/'
2026-07-09T14:23:12.483+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Started SpringLearnApplication in 3.678 seconds (process running for 4.642)
2026-07-09T14:23:12.561+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Inside main
2026-07-09T14:23:12.564+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Start
2026-07-09T14:23:12.564+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Inside Country Constructor.
2026-07-09T14:23:12.567+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Inside setCode()
2026-07-09T14:23:12.567+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Inside setName()
2026-07-09T14:23:12.568+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Country : Country [code=IN, name=India]
2026-07-09T14:23:12.564+05:30 INFO 06988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : End
```

Hello World RESTful Web Service

```

:: Spring Boot ::      (v3.5.16)

2026-07-10T19:46:42.182+05:30 INFO 21988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.11 with PID 21988 (C:\Users\shiva\Down
2026-07-10T19:46:42.188+05:30 INFO 21988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Running with Spring Boot v3.5.16, Spring v6.2.19
2026-07-10T19:46:42.190+05:30 INFO 21988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-10T19:46:42.325+05:30 INFO 21988 --- [spring-learn] [restartedMain] e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to
2026-07-10T19:46:42.325+05:30 INFO 21988 --- [spring-learn] [restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property
2026-07-10T19:46:44.934+05:30 INFO 21988 --- [spring-learn] [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-10T19:46:44.971+05:30 INFO 21988 --- [spring-learn] [restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-10T19:46:44.972+05:30 INFO 21988 --- [spring-learn] [restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.55]
2026-07-10T19:46:45.077+05:30 INFO 21988 --- [spring-learn] [restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-10T19:46:45.081+05:30 INFO 21988 --- [spring-learn] [restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2752 ms
2026-07-10T19:46:46.030+05:30 INFO 21988 --- [spring-learn] [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : LiveReload server is running on port 35729
2026-07-10T19:46:46.142+05:30 INFO 21988 --- [spring-learn] [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-10T19:46:46.167+05:30 INFO 21988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Started SpringLearnApplication in 5.293 seconds (process running for 6.545)
2026-07-10T19:46:46.175+05:30 INFO 21988 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Inside main
```